

Problem 5

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Problem 1 An object moving along a straight line has a position given by $s(t) = \frac{1}{t-4}$, where s is measured in meters and t in seconds. Find the velocity of the object at time $t = 6$.

Explanation. $v(6) = s'(6) = \lim_{t \rightarrow 6} \frac{s(t) - s(6)}{t - 6}$. When we check the form of this

$$\begin{aligned} \text{limit, we see that it has form } \frac{0}{0}. \quad s'(6) &= \lim_{t \rightarrow 6} \frac{\frac{1}{t-4} - \frac{1}{2}}{t-6} = \lim_{t \rightarrow 6} \frac{\frac{2 - (t-4)}{2(t-4)}}{t-6} = \\ &= \lim_{t \rightarrow 6} \frac{(2-t+4)}{2(t-4)(t-6)} = \\ &= \lim_{t \rightarrow 6} \frac{(6-t)}{2(t-4)(t-6)} = \lim_{t \rightarrow 6} \frac{-1}{2(t-4)} = \frac{-1}{2(6-4)} = \frac{-1}{4} \text{ m/s.} \end{aligned}$$